

No. of Functional Features Included in the Solution

Date	02 June 2026
Team ID	XXXXXX
Project Name	Raising waters
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Interface	Allows users to enter weather parameters such as rainfall and cloud visibility.	Flask Web Application	Done	High
2	Data Preprocessing	Cleans and prepares historical weather data for prediction.	Data Processing	Done	High
3	Machine Learning Model	Trains Decision Tree, Random Forest, KNN, and XGBoost models for flood prediction.	ML Module	Done	High
4	Flood Risk Prediction	Predicts whether flood conditions are likely based on user inputs.	Prediction Engine	Done	High
5	Model Evaluation	Compares multiple algorithms and selects the best-performing model (XGBoost – 96.55% accuracy).	Model Evaluation	Done	Medium
6	Prediction Results Display	Displays prediction results through the Flask web interface.	User Interface	Done	Medium
7	IBM Cloud Deployment Support	Prepares the application for deployment on IBM Cloud.	Deployment	Done	Medium
8	Real-Time Weather API Integration	Fetches live weather data automatically for prediction.	API Integration	Pending	Low

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Weather Input Form, Prediction Result Page
2	Backend / Logic	3	Data Preprocessing, ML Prediction, Model Evaluation
3	Database / Storage	1	CSV Dataset, Trained Model (.pkl)
4	API / Integration	1	IBM Cloud Deployment / Future Weather API
5	Security / Authentication	1	Basic Input Validation